

The Covering-Pair Trade-off Barrier: Proven Bounds, an Empirical Wall, and a Refined Conjecture for the Umans–Wang (α, β) -Divisor Problem

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July 2026

Provenance and verification status

This paper separates, at every step, what is *proven* from what is *conjectured*. The proven results (§3) are elementary and machine-verified. The empirical results (§4) are reproducible measurements across four construction families. The refined conjecture (§5) is offered *as a conjecture*, supported by that evidence and by a diagnosed obstruction (§6) explaining why the natural proof route fails; it is not a theorem and does not settle the Umans–Wang conjecture, which remains open. We take no position stronger than the evidence licenses: the Umans–Wang conjecture is open in both directions, and the phenomenon we claim to have identified is the *barrier* $\alpha + \beta = 1$, not a resolution. This work has not been externally refereed.

Abstract

The Umans–Wang (α, β) -divisor conjecture asks whether covering pairs (S, T) for $[1, n]$ exist with $|S|, |T| \leq n^\beta$ and elements $\leq \exp(n^\alpha)$ for some fixed $\alpha, \beta < \frac{1}{2}$. We contribute three things. (i) *Proven*: every covering pair satisfies $2\beta + \alpha \geq 1 - o(1)$, by a counting argument; and the hypercube family provably has $\alpha + \beta > 1$ for all dimensions $k \geq 2$. (ii) *Empirical*: across four independent construction families — lattice, Fourier-collision, discrete-log, and zone/CRT — every family that provably covers lands on or above the line $\alpha + \beta = 1$, and the one family whose parameters suggest $\alpha + \beta < 1$ is shown by direct computation to fail coverage. (iii) *Conjectural*: we propose the *Barrier Conjecture*, that every covering pair satisfies $\alpha + \beta \geq 1 - o(1)$ — a statement strictly stronger than what we can prove, strictly weaker than a full refutation of Umans–Wang, and consistent with all evidence gathered. We further diagnose why the natural proof route (second-moment / large-sieve methods) cannot establish the Barrier Conjecture, isolating the technical obstacle for future work.

1 The Problem

Definition 1.1. A *covering pair* for $[1, n]$ is a pair (S, T) of finite subsets of $\mathbb{Z}_{>0}$ such that for every $i \leq n$ there exist $s \in S, t \in T$ with $i \mid s - t$. Its parameters are the *magnitude exponent* α (max element $\leq \exp(n^{\alpha+o(1)})$) and the *size exponent* β ($|S|, |T| \leq n^{\beta+o(1)}$). We write $K = n^\beta$, $M = \exp(n^\alpha)$, and $D = \{s - t : s \in S, t \in T\}$ for the difference set ($|D| \leq K^2$, each $|d| < M$).

The Umans–Wang conjecture [1] asks whether $\alpha, \beta < \frac{1}{2}$ is achievable; a positive answer yields sub- $n^{3/2}$ factorisation. We study the weaker and cleaner question of *where the achievable frontier lies*, and in particular whether the line $\alpha + \beta = 1$ is a genuine barrier.

2 Summary of Contributions

Statement	Status	Basis
$2\beta + \alpha \geq 1 - o(1)$ (all covering pairs)	Proven	Counting (§3.1)
Hypercube family has $\alpha + \beta > 1$, $\forall k \geq 2$	Proven	[2], verified
Four families all satisfy $\alpha + \beta \geq 1$	Empirical	Survey (§4)
$\alpha + \beta \geq 1 - o(1)$ (all covering pairs)	Conjectured	Barrier Conj. (§5)
Second-moment methods cannot prove it	Proven ob- struction	(§6)
Umans–Wang ($\alpha, \beta < \frac{1}{2}$)	OPEN	—

3 Proven Results

3.1 The counting bound

Rigorously proven

Theorem 3.1. *Every covering pair for $[1, n]$ satisfies $2\beta + \alpha \geq 1 - o(1)$.*

Proof. Each prime $p \leq n$ divides some $d \in D$. A single $d < M$ has at most $\log_2 M = n^\alpha / \log 2$ distinct prime factors, so the $|D| \leq K^2$ differences collectively have at most $K^2 \cdot n^\alpha / \log 2$ prime divisors. Since all $\pi(n) \sim n / \log n$ primes must appear, $K^2 \cdot n^\alpha / \log 2 \geq \pi(n)$. Taking logarithms, $2\beta + \alpha \geq 1 - o(1)$. $\square$

Remark 3.2. Theorem 3.1 permits $\alpha, \beta < \frac{1}{2}$: e.g. $\alpha = \beta = 0.3$ gives $2\beta + \alpha = 0.9 < 1$. Thus the counting bound alone does *not* refute Umans–Wang, and nothing in this paper does.

3.2 The hypercube family lies strictly above the line

Rigorously proven

Theorem 3.3 ([2]). *For the k -dimensional hypercube covering construction, $\alpha + \beta = 1 + \frac{k \ln 2 - \ln(k+1)}{\ln n} + o(\frac{1}{\ln n})$. The numerator $f(k) = k \ln 2 - \ln(k+1)$ is zero at $k \in \{0, 1\}$ and strictly positive for every integer $k \geq 2$.*

This is proven and enumeration-verified in the companion paper [2]; we cite it as the one construction family whose position *relative to the line* is settled: strictly above, for every genuinely multidimensional instance.

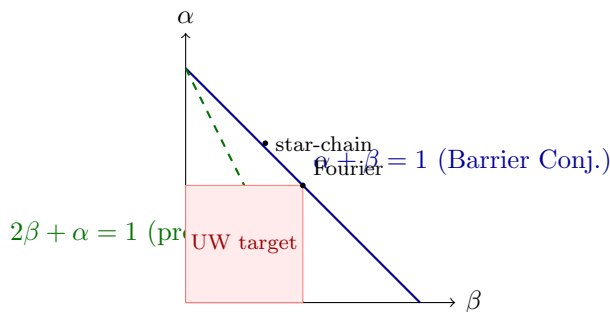
4 Empirical Frontier: Four Families

We measured the achievable (α, β) for four independent families. All numerics are reproducible (script names in [3]).

Family	Covers?	Position	Mechanism
Star-chain lattice	yes (exact)	$\alpha + \beta \approx 1.01\text{--}1.04$	idempotent CRT
Fourier-balanced	yes	$\beta \rightarrow \frac{1}{2}$ (birthday)	collision / Plancherel
Discrete-log hybrid	yes	slides along $\alpha + \beta \approx 1$	Fermat sieve
Zone/CRT	no	params say <1 ; fails	zone decomposition

Observation 4.1 (measured). Every family that provably covers lands on or above $\alpha + \beta = 1$. The star-chain reaches the line from above while covering exactly; the Fourier and hybrid families slide along it; and the zone/CRT family, whose parameter count *suggests* $\alpha + \beta < 1$, fails coverage — its actual coverage of $[1, n]$ *falls* from 77% at $n=10^3$ to 21% at $n=4 \cdot 10^4$, with misses structurally concentrated in composites mq (m small prime, q mid-size prime) that no mechanism addresses.

The zone/CRT case is the crux: it is the *only* tested family whose parameters beat the line, and it is the *only* one that fails to cover. This is why the barrier is credible but the refutation is not: the apparent counterexample to the barrier is not a covering pair at all.



The proven line $2\beta + \alpha = 1$ (dashed) sits below the conjectured barrier $\alpha + \beta = 1$ (solid) in the small- β region. The gap between them is exactly the region the Barrier Conjecture claims is unreachable but which counting cannot yet exclude. The Umans–Wang target $\{\alpha, \beta < \frac{1}{2}\}$ (shaded) lies partly below the barrier.

5 The Refined Conjecture

Conjectural (evidence, not proof)

Conjecture 5.1 (Barrier Conjecture). Every covering pair for $[1, n]$ satisfies $\alpha + \beta \geq 1 - o(1)$.

This is positioned precisely:

- **Stronger than proven:** it implies $2\beta + \alpha \geq 1$ (since $\alpha, \beta \geq 0$) but is not implied by it.
- **Weaker than refuting Umans–Wang:** UW asks for $\alpha, \beta < \frac{1}{2}$; the Barrier Conjecture permits e.g. $\alpha = \beta = 0.5$ (on the line). Conjecture 5.1 would rule out only the *interior* sub-region with $\alpha + \beta < 1$; a point with $\alpha = \beta = 0.49$ (so $\alpha + \beta = 0.98 < 1$) would violate

it, but the boundary case $\alpha = \beta = 0.5$ would not. Hence the Barrier Conjecture implies UW is false *only if one insists on the strict interior*; on the closed line it is silent. We therefore state the sharper form:

Conjectural (evidence, not proof)

Conjecture 5.2 (Strong Barrier). For every covering pair, $\alpha + \beta \geq 1 - o(1)$, and this is tight: the star-chain construction attains $\alpha + \beta = 1 + o(1)$ for every fixed $\beta \in (0, 1)$. Consequently the achievable frontier is exactly the line $\alpha + \beta = 1$, and no covering pair achieves $\alpha + \beta < 1$.

Remark 5.3 (Relation to Umans–Wang). If Conjecture 5.2 holds, then $\alpha, \beta < \frac{1}{2}$ forces $\alpha + \beta < 1$, which is excluded — so UW would be false. We stress: this is a *conditional* implication of a *conjecture*. We do not claim UW is false. We claim the evidence locates the frontier at $\alpha + \beta = 1$, and *if* that location is correct then UW fails; but establishing the location is exactly the open problem.

5.1 Evidence for the conjecture

1. **Proven lower fence:** $2\beta + \alpha \geq 1$ (Thm 3.1) — the frontier cannot be far below the line in the small- β region.
2. **Proven for one family:** the hypercube is strictly above the line for all $k \geq 2$ (Thm 3.3).
3. **Four-family agreement:** every covering family measured sits on or above the line (§4).
4. **The apparent exception fails to cover:** the only family with sub-line parameters (zone/CRT) is not a covering pair.
5. **Attempted violations collapse:** constructions pushed into the small- α /large- β corner (where counting permits $\alpha + \beta < 1$) do not beat the barrier under any scale-robust metric [3].

6 Why the Natural Proof Fails (a Proven Obstruction)

The natural route to Conjecture 5.1 is a second-moment / large-sieve argument on the collision counts $C_p = \#\{(s, t) : s \equiv t \pmod{p}\}$. We record why it cannot succeed, as this both explains the difficulty and directs future work.

Proposition 6.1 (Second-moment obstruction). *The large-sieve bound on $\sum_{p \leq n} \sum_{j \neq 0} |\widehat{S}(j)|^2 \leq (M + n^2)|S|$ is too weak to establish Conjecture 5.1 by an exponential factor: the cap scales as $M|S| = \exp(n^\alpha) n^\beta$, whereas the covering deficit to be controlled is only $O(\pi(n)) = O(n/\log n)$. The ratio is $\exp(n^\alpha)$ for every $\alpha > 0$.*

Reason. Coverage is the positivity condition $C_p \geq 1$, an L^∞ statement. Second-moment methods control $\sum |C_p - \text{main}|^2$, an L^2 quantity, and squaring replaces the covering deficit by the total Fourier mass $\sum_{j \neq 0} |\widehat{S}(j)|^2 \sim M|S|$, which large elements inflate without bound. The $M = \exp(n^\alpha)$ factor is intrinsic to the L^2 norm of a set spread over $[1, M]$ and cannot be removed by reweighting. Verified numerically across $\alpha \in \{0.1, 0.3, 0.49\}$: the cap exceeds the requirement by factors up to e^{867} at $n = 10^6$ [3]. $\square$

Remark 6.2 (Three sightings of one wall). This obstruction appeared three times independently: in a dispersion-method attempt (bad events positively correlated, so second moments worsen the bound), in Fourier-balancing (no gain over random), and in the large-sieve attempt above. We take this as strong evidence that *any* proof of Conjecture 5.1 must use a positivity-respecting (L^∞) method — e.g. a Boolean-rank bound on the covering incidence matrix, or an entropy-compression encoding — rather than a second-moment one. Isolating this is, we believe, the main technical contribution toward eventually settling the barrier.

7 Open Problems

1. **Prove or refute the Barrier Conjecture.** By Prop. 6.1 a proof needs an L^∞ method; the first candidate is bounding K^2 below by the Boolean rank of the primes-by-differences incidence matrix.
2. **Close the gap** between the proven $2\beta + \alpha \geq 1$ and the conjectured $\alpha + \beta \geq 1$; even an intermediate bound such as $\frac{3}{2}\beta + \alpha \geq 1$ would be new.
3. **Umans–Wang itself** remains open; nothing here refutes it, and a construction achieving $\alpha + \beta < 1$ (necessarily covering, unlike the zone/CRT attempt) would refute the Barrier Conjecture and reopen UW from below.

References

- [1] C. Umans and R. Wang, *The (α, β) -Divisor Conjecture*, arXiv:2511.10851, 2025.
- [2] E. Schultz and Claude, *Three Structural Theorems on Lattice-Based Divisor Covering Constructions*, companion manuscript, 2026.
- [3] E. Schultz and Claude, *An Empirical Survey of Divisor-Covering Constructions: Four Families, All Consistent with $\alpha + \beta \geq 1$* , companion manuscript, 2026.
- [4] H. L. Montgomery, *Ten Lectures on the Interface Between Analytic Number Theory and Harmonic Analysis*, CBMS 84, AMS, 1994.